Corrected MCQs

1. **C.** The quotient tends to 0, so the sum tends to 3. Neither the numerator nor the first term Give the limit.
2. **A.** Divide numerator and denominator by n^2. The dominant terms are of the same degree.
3. **B.** The product tends to 2 $\times 0 = 0$. The quotient may be undefined, or change a sign; the sum tend to 2.
4. **D.** $3/(n+1) < 1/10$ is equivalent to $n > 29$. In row 29, the difference is exactly 0.1.
5. **B.** The same terms are worth 1, the odd -1 ; they cannot stay in the same small band The sequence is bounded.
6. **C.** Monotonous convergence. Constant sequences equal to -1 refute positivity and limit 5; $5 - 1/(n+1)$ Refuses constancy.
7. **D.** $-1/n \leq u_n \leq 1/n$, and both bounds tend towards 0; no monotonicity is required.
8. **A.** For every threshold, a term exceeds it, as do all later terms since the sequence is increasing. This is the definition limit $+\infty$.
9. **C.** By conjugation, the expression is 1 $/(\sqrt{1+1/n}+1)$ for $n > 0$, where $1/2$. Subtraction Infinites did not allow to conclude.
10. **B.** The upper bound tends towards 0. A sequence between the bounds can go up on time, e.g. by cancelling the same terms.

True/false. False: $(-1)^n/(n+1) \to 0$ is not monotonous. True by monotonous convergence. False: $5 - 1/(n+1) \to 5$; only wide inequality passes to the limit.

Corrected applications and support

Application 1. The difference at 5 is $2/(n+1)$; we need $n+1>40$, so $n \geq 40$. In row 39, $v_{39} = 4{,}95$ is Excluded. Variations decrease, resulting in validity for all subsequent ranks.

Application 2. The following has terms equal to 0 in any band, but for $\varepsilon = 1/2$, any row is followed by an odd row whose term is 1. N ensures the universal condition.

Implementation 3. The first quotient tends to 0 by division by n^2. The second expression is $\frac{n}{\sqrt{n^2+n}+n} = \frac{1}{\sqrt{1+1/n}+1} \to 1/2$. The third has an absolute value equal to 1 $/(n+1) \to 0$, so tend towards 0.

Implementation 4. $a_n = 2^n \to +\infty$. Equation $\ell = 2\ell$ is a necessary condition if a finite limit exists; it does not prove this existence.

Application 5. Initially $b_0 = 0$. If $0 \leq b_n < 3$, then $2 \leq b_{n+1} < 3$, therefore the invariant is transmitted. $b_{n+1} - b_n = \frac{2}{3}(3 - b_n) > 0$. Monotonous convergence; the formula $b_n = 3 - 3(1/3)^n$ gives the limit 3. Out of continuity, $\ell = \ell/3 + 2$ also gives 3 after establishing convergence.

Implementation 6. $(-1)^n$ is bounded between -1 and 1, not monotonous and not converging because its sub-sequences pairs and odds have different limits. $-1/(n+1)$ is increasing, bounded between -1 and 0, and converging towards 0. n is increasing, not increased and diverges to $+\infty$.

Support points

Color the words that remain outside the band, then translate "remain" with the quantifiers.
Prior to S32, give continuity as an acceptable result in an identification exercise
or use an explicit formula. Proof of monotonous convergence itself remains accepted;
its implementation must be reusable.

Conduct and correction

Benchmarks for the teacher

The DS2 precedes S19 and S20: assess definitions and operations already encountered, without require monotonous convergence, Bernoulli or gendarmes. Draw horizontal bands around the limit and search for a row from which all the terms remain. convergence before solving the fixed point equation. Show $(-1)^n$ for dissociating "bounded" and "converging".

Corrected activity

1. $u_0 = 5$, $u_9 = 2.3$ and $u_{99} = 2.03$. Since $\frac{3}{n+1} > 0$ at any level, $u_n > 2$: no term equals 2.

2. $|u_n - 2| < 0{,}01 \iff \frac{3}{n+1} < 0{,}01 \iff n+1 > 300 \iff n \geq 300$. The first appropriate ranking is so $N = 300$. At range 299, the gap is exactly 0.01 and strict inequality is not satisfied.

3. For $\varepsilon > 0$, let's take it $N = \lceil \frac{3}{\varepsilon} \rceil$ the smallest integer greater than or equal to $\frac{3}{\varepsilon}$. If $n \geq N$, then $n+1 > \frac{3}{\varepsilon}$, so $|u_n - 2| < \varepsilon$. This choice is appropriate, even when it is not minimal.

4. $|v_n - 2| = \frac{1}{n+1} \to 0$, therefore $v_n \to 2$. But $v_0 = 3$, $v_1 = \frac{3}{2}$ and $v_2 = \frac{7}{3}$: one sees a decrease and then a This converging sequence is not monotonous.

E1. For $n \geq 1$, $\frac{4n-1}{n+2} = \frac{4-\frac{1}{n}}{1+\frac{2}{n}} \to 4$, because the denominator tends to $1 \neq 0$. Similarly $n^2 - 7n = n^2(1 - \frac{7}{n})$; the first factor tends to $+\infty$ and the second about $1 > 0$, so the product tends to $+\infty$.

E2. The conjugate gives $\sqrt{n+1} - \sqrt{n} = \frac{(n+1)-n}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}}$. The denominator tends to $+\infty$, therefore the difference tends to 0.

E3. For $n \geq 1$, $-1 \leq \cos n \leq 1$ and $n > 0$, therefore $-\frac{1}{n} \leq u_n \leq \frac{1}{n}$. The two bounds tend towards 0; the theorem of the gendarmes gives $u_n \to 0$.

E4. The continuation is increasing and increased by 5, so converge. By passing to the limit in $u_n \leq 5$, its limit check $\ell \leq 5$. It is not necessarily equal to 5: $u_n = 2 - \frac{1}{n+1}$ is increasing, increased by 5 and converging to 2.

E5. Like $|-0{,}6| < 1$, $(-0{,}6)^n \to 0$, therefore $u_n \to 3$. However $u_0 = 1$, $u_1 = 4{,}2$ and $u_2 = 2{,}28$: Then go up and down and is not monotonous.

Corrected from DM2

1. Note: $P(n)$: « u_n is defined and $0 \leq u_n \leq 2$". This is true in row 0 since $u_0 = 0$. Fixes $n \geq 0$ and Suppose $P(n)$ We've got two $\leq 2 + u_n \leq 4$, so $u_{n+1} = \sqrt{2+u_n}$ is defined and $\sqrt{2} \leq u_{n+1} \leq 2$. So $P(n+1)$ is true. By induction, all the terms belong to $[0,2]$.

2. Poisons $R(n)$: « $u_{n+1} \geq u_n$ For $n = 0$, $u_1 = \sqrt{2} \geq 0 = u_0$. Fixing $n \geq 0$ and suppose $R(n)$ The growth of $x \mapsto \sqrt{2+x}$ on $[0,2]$ gives

$$u_{n+2} = \sqrt{2+u_{n+1}} \geq \sqrt{2+u_n} = u_{n+1}.$$

So $R(n+1)$ holds. By induction, (u_n) is increasing. It is increased by 2, so converge to a real $\ell \in [0,2]$.

3. The function $x \mapsto \sqrt{2+x}$ is continuous in ℓ. By passing to the limit, $\ell = \sqrt{2+\ell}$. Like $\ell \geq 0$, this is equivalent to $\ell^2 - \ell - 2 = 0$, either $(\ell - 2)(\ell + 1) = 0$. Root -1 is excluded; therefore $\ell = 2$.

4. Using the conjugate and $u_n \leq 2$,

$$0 \leq 2 - u_{n+1} = \frac{2 - u_n}{2 + \sqrt{2 + u_n}} \leq \frac{2 - u_n}{2}.$$

Note: $T(n)$: « $0 \leq 2 - u_n \leq \frac{2}{2^n}$ In row 0, 2 $- u_0 = 2$. If $T(n)$ is true for a fixed integer $n \geq 0$, then

$$0 \leq 2 - u_{n+1} \leq \frac{1}{2}\frac{2}{2^n} = \frac{2}{2^{n+1}}.$$

Initialisation and inheritance prove $T(n)$ Any way. $\frac{2}{2^{11}} = \frac{1}{1024} < 10^{-3}$, $n = 11$ is enough.
It is a rank guaranteed by an increase, without a statement of minimality.
Before question 3: Number 2 is already involved in the framework.

Corrected work in the manual or an anal

Fiscal year 17. The quotients are first of indeterminate form. $n \geq 1$,

$$u_n = \frac{2n - 4}{n^2 + 1} = \frac{2\left(1 - \frac{2}{n}\right)}{n\left(1 + \frac{1}{n^2}\right)} \longrightarrow 0.$$

Similarly,

$$v_n = \frac{2 - 5n}{4n + 7} = \frac{\frac{2}{n} - 5}{4 + \frac{7}{n}} \longrightarrow -\frac{5}{4}, \qquad w_n = n\frac{-1 - \frac{10}{n^2} + \frac{4}{n^3}}{2 + \frac{3}{n} + \frac{1}{n^2}} \longrightarrow -\infty.$$

The last quotient tends to $-\frac{1}{2}$, and factor n To $+\infty$.

Fiscal year 24. For every $n \geq 1$, $-1 \leq (-1)^n \leq 1$. Since $n^2 > 0$, we get $-\frac{1}{n^2} \leq u_n \leq \frac{1}{n^2}$. Both limits tend towards 0; the theorem of gendarmes gives $u_n \to 0$.